Appendix

Numerical Analysis

Numerical Integration

A Resonance-free Adaptive Levin Method in Two-Dimensions

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$$\int_R f(\mathbf{x}) \; \exp(ig(\mathbf{x})) \; d\mathbf{x}$$

Adaptive Integration

- Deploy Levin Method
- Outperforms all methods in high frequency regime
- Time complexity is independent of frequency

ODE Solvers

A SOLVER FOR LINEAR SCALAR ORDINARY DIFFERENTIAL EQUATIONS WHOSE RUNNING TIME IS BOUNDED INDEPENDENT OF FREQUENCY

MURDOCK AUBRY* AND JAMES BREMER†

$$p(t)y(t) + \sum_{i=1}^{N} q_i(t)\frac{\partial^i y}{\partial t^i} = 0$$

Numerical solver for large order scalar equations

- Generalizes to all dimensions
- Outperforms all methods in high frequency regime
- Time complexity is independent of frequency